Determinants of Bilharzia Prevalence in the Coastal Region of Kenya Among Children Aged 9–15 years using Spatial Statistical Analysis

SCT-254-103/2022 OTIENO JAMES OJUNGO

SUPERVISOR: MR. HENRY MWANGI

A research project submitted to the Department of Mathematics in partial fulfillment of the requirements for the award of the degree of Bachelor of Science in Mathematics and Computer Science, Multimedia University of Kenya.

2025

Dedication

I dedicate this work to my family for their unwavering support, encouragement, and inspiration throughout my academic journey. Their patience and belief in my abilities have been a constant source of motivation during the preparation of this research project.

Declaration

This research project is my original work and has not been presented for the award of a degree in any other university.

James Otieno sign: 15th /06/2026

This project research has been submitted for examination with my approval as the University Supervisor.

Henry Mwangi sign: 16th/06/2026

Acknowledgements

I would like to express my sincere gratitude to the supervisor for the valuable guidance, constructive criticism, and continuous support throughout the development of this research project. His guidance largely contributed to shaping this study. I also extend my appreciation to the Department of Mathematics at Multimedia University of Kenya for providing an enabling academic environment that facilitated this work. Furthermore, I am deeply grateful to my classmates and families for their encouragement and moral support during the preparation of this project.

Table of Contents

Dedication i

Acknowledgements iii

Background Information 1

Statement Of the Problem 2

Justification Of the Study 3

Objectives Of the Study 4

General Objective 4

Specific Objectives 4

Research Questions 4

Hypotheses Of the Study 4

Scope Of the Study 5

Definitions of Key Terms 5

CHAPTER TWO: LITERATURE REVIEW 7

Introduction 7

Schistosomiasis 7

Global and Regional Burden of Schistosomiasis 8

Schistosomiasis Among Children Aged 9–15 Years 8

Determinants Of Bilharzia Prevalence 9

Environmental Determinants 9

Demographic Determinants 10

Control Strategies 10

Mass Drug Administration 10

Spatial Epidemiology and Statistical Approaches 11

Research Gaps 11

Summary 12

CHAPTER THREE: METHODOLOGY 13

Introduction 13

Study Area 13

Study Design 13

Study Population 14

Data Sources 14

Variables Of the Study 14

Dependent Variable 14

Independent Variables 14

Data Management 15

Data Analysis 15

Spatial Descriptive Analysis 15

Spatial Autocorrelation Analysis 16

Spatial Association Analysis 17

Software Used 18

Ethical Considerations 18

CHAPTER FOUR: DATA ANALYSIS 19

Data Cleaning 19

Overall Bilharzia Prevalence 19

Spatial Variation in Bilharzia Prevalence Across Sub-counties (objective 1) 19

Statistical Test for Spatial Variation (objective 2) 21

Spatial Distribution of Demographic Characteristics and Association with Prevalence (objective 3) 22

Individual Level Association 23

Gender 23

Age 23

Aggregate Level Associations 24

Spatial Association Between Environmental Factors and Prevalence (objective 4) 24

Individual-Level Associations 26

Aggregate Level Associations 26

Presence and Strength of Spatial Autocorrelation in Bilharzia Prevalence (objective 4) 26

Multivariate Risk Models 27

Multivariate Logistic Regression 27

Multivariate Linear Regression 28

Interpretation of Risk Factors 29

CHAPTER FIVE: INTERPRETATION, CONCLUSION AND RECOMMENDATIONS 30

Interpretation of Key Findings 30

Spatial Variation and Clustering (Objective 1 & 4) 30

Demographic Risk Factors (Objective 2) 30

Environmental Risk Factors and Multivariate Insights (Objective 3) 31

Implications for Interventions 32

Interventions with County and National Authorities 32

Interventions Involving Residents and Communities 33

Integrated Intervention Framework 34

Study Limitations 35

Recommended Future Research: 35

Conclusion and Policy Recommendations 35

References 37

Appendices 40

Work plan 40

Budget 41

CHAPTER ONE: INTRODUCTION

Background Information

Bilharzia (schistosomiasis) is a tropical disease caused by parasitic worms of the genus *Schistosoma*. It remains a major public health concern in sub-Saharan Africa, including Kenya, where over 2.5 million people are at risk of infection. The disease occurs in two main forms: *Schistosoma haematobium* that causes urogenital schistosomiasis, and *Schistosoma mansoni* that causes intestinal schistosomiasis. Both forms are transmitted through contact with freshwater bodies infested with infective cercariae released by intermediate snail hosts (Chacha G., 2025).

In Kenya, Bilharzia prevalence shows clear geographical variation, with higher rates in areas where ecological and hydrological conditions favor snail habitats and where communities frequently interact with freshwater sources. The Coastal region of Kenya, characterized by numerous rivers, ponds, and water bodies used for domestic and recreational purposes, continues to experience persistent transmission of Bilharzia despite ongoing control measures. Understanding how prevalence varies spatially is crucial for designing effective interventions (Kepha et al., 2023).

Spatial epidemiology has demonstrated that schistosomiasis does not occur randomly across space but tends to cluster in high-risk areas, often associated with environmental, demographic, and socio-behavioral factors (Mutono et al., 2025). Precision mapping studies in the Kenya Coastal region have documented marked variation in schistosomiasis prevalence at sub-county and ward levels, with overall prevalence estimates averaging about 6.9% among school-age children, but ranging as high as nearly 20% in certain counties such as Tana River (Kepha et al., 2023). Such spatial differences highlight the need to use spatial statistical methods

such as choropleth mapping and spatial autocorrelation to identify and quantify geographical patterns of infection.

Children aged 9–15 years are considered a key demographic for schistosomiasis studies because they often engage in activities involving water contact which increases their risk of exposure, for instance, swimming, fishing, and washing. These patterns can differ significantly by location, age, and gender, making spatially stratified analysis essential for identifying high-risk areas and understanding the determinants of infection (Kepha et al., 2023).

Spatial statistics methods that explicitly account for the geographical location of observations provide powerful tools to reveal where disease prevalence is highest, whether there is clustering, and how environmental and demographic factors vary across space. By aggregating data at administrative levels such as sub-counties and wards, spatial analysis enables targeted public health planning and resource allocation, improving the effectiveness of interventions such as mass drug administration (MDA) and improved water, sanitation, and hygiene services (Kepha et al., 2023).

Statement Of the Problem

Despite national efforts against Bilharzia, including periodic mass drug administration and health education campaigns, the disease persists among children aged 9–15 years in the Coastal region. However, existing studies have predominantly focused on overall prevalence without fully examining spatial distribution and clustering patterns across smaller administrative units such as sub-counties or wards.

Spatial heterogeneity in infection rates suggests that some areas experience significantly higher prevalence due to local environmental conditions, water usage patterns, and access to sanitation facilities, while others may have lower rates. Traditional non-spatial statistical

approaches may not effectively capture such clustering and locational dependence, leading to sub-optimal targeting of control measures. Therefore, there is a need for spatial statistical analysis that can identify high-prevalence areas and quantify spatial associations between environmental and demographic factors and Bilharzia prevalence among children aged 9–15 years.

Justification Of the Study

This study is justified on both scientific and public health grounds. Bilharzia prevalence is not uniformly distributed across geographical space, and evidence from precision mapping studies in the Coastal region of Kenya indicates that infection levels vary widely between sub-counties and wards, with some areas experiencing significantly higher disease burdens than others. Such spatial variation highlights the importance of understanding geographical patterns of infection rather than relying solely on overall prevalence estimates. Furthermore, identifying geographic clustering of Bilharzia prevalence supports more effective and targeted public health interventions, enabling health authorities to concentrate limited resources, such as mass drug administration and sanitation improvements, in areas where they are most needed. The study also focuses on school-age children, who experience the highest exposure and disease burden due to frequent contact with contaminated water sources, making them a priority group for both research and intervention. In addition, findings from spatial analysis can be integrated into national and county health planning processes, aligning with Kenya’s strategies for the control of neglected tropical diseases and informing future disease surveillance and control efforts.

Objectives Of the Study

General Objective

To examine the spatial distribution and spatial determinants of Bilharzia prevalence among children aged 9–15 years in the Coastal region of Kenya.

Specific Objectives

1. To analyze the spatial variation in Bilharzia prevalence across sub-counties in the Coastal region.
2. To examine the spatial distribution of demographic characteristics (age and gender) and their association with Bilharzia prevalence across sub-counties.
3. To assess the spatial association between environmental factors (water source type and toilet availability) and Bilharzia prevalence across sub-counties.
4. To assess the presence and strength of spatial autocorrelation in Bilharzia prevalence among sub-counties in the Coastal region.

Research Questions

1. How does Bilharzia prevalence vary spatially across sub-counties in the Coastal region of Kenya?
2. How are aggregated demographic characteristics (age and gender) spatially associated with Bilharzia prevalence across sub-counties?
3. What is the spatial association between environmental factors (type of water source and toilet availability) and Bilharzia prevalence across sub-counties?
4. Is there significant spatial autocorrelation in Bilharzia prevalence across sub-counties in the Coastal region?

Hypotheses Of the Study

1. H_{01}: There is no significant spatial variation in Bilharzia prevalence across sub-counties in the Coastal region of Kenya.
2. H_{11}: There is significant spatial variation in Bilharzia prevalence across sub-counties in the Coastal region of Kenya.
3. H_{02}: There is no significant spatial association between aggregated demographic characteristics (age and gender distribution) and Bilharzia prevalence across sub-counties.
4. H_{22}: There is a significant spatial association between aggregated demographic characteristics (age and gender distribution) and Bilharzia prevalence across sub-counties.
5. H_{03}: There is no significant spatial association between environmental factors (water source type and toilet availability) and Bilharzia prevalence across sub-counties.
6. $H_{33:}$ There is a significant spatial association between environmental factors (water source type and toilet availability) and Bilharzia prevalence across sub-counties.
7. H_{04}: Bilharzia prevalence across sub-counties does not exhibit significant spatial autocorrelation.
8. $H_{44:}$ Bilharzia prevalence across sub-counties exhibits significant spatial autocorrelation.

Scope Of the Study

The study focuses on the Coastal region of Kenya, encompassing all sub-counties with available data on Bilharzia infection among children aged 9–15 years. Spatial analysis is conducted at the sub-county level using aggregated demographic and environmental variables linked to geographical shapefiles.

Definitions of Key Terms

- Spatial statistics: Statistical methods that explicitly analyze data with reference to geographical location.

- Bilharzia: A parasitic disease caused by *Schistosoma* species.
- Prevalence: Proportion of individuals infected in a population at a given time.
- Spatial autocorrelation: The degree to which similar values cluster in geographical space.

CHAPTER TWO: LITERATURE REVIEW

Introduction

This chapter reviews existing literature on schistosomiasis (Bilharzia), focusing on its prevalence among school-aged children, determinants of infection, and spatial distribution patterns. The review draws on global, regional, and Kenyan empirical studies to examine how environmental and demographic factors influence Bilharzia transmission. Particular emphasis is placed on spatial epidemiology and spatial statistical approaches, which provide a framework for understanding geographical heterogeneity in disease burden. This chapter provides the theoretical and empirical foundation for the current study, which investigates the spatial distribution and determinants of Bilharzia prevalence among children aged 9–15 years in the Coastal region of Kenya.

Schistosomiasis

Schistosomiasis is a parasitic disease caused by trematode worms of the genus *Schistosoma*. Human infection occurs when the skin comes into contact with freshwater contaminated with cercariae released by infected freshwater snails, which serve as intermediate hosts (WHO, 2023). Once inside the human body, the parasites mature and lay eggs, some of which are excreted while others become trapped in body tissues, causing inflammation and long-term morbidity.

Transmission of schistosomiasis is closely linked to environmental and behavioral factors. Communities that rely on rivers, lakes, irrigation canals, and ponds for domestic, recreational, or occupational activities are at increased risk of infection. Poor sanitation further exacerbates transmission by allowing parasite eggs to contaminate freshwater bodies. Due to

these characteristics, schistosomiasis exhibits strong geographical variation, making it particularly suitable for spatial analysis (Colley et al., 2014).

Global and Regional Burden of Schistosomiasis

Schistosomiasis remains one of the most widespread neglected tropical diseases globally. The World Health Organization estimates that more than 250 million people required preventive chemotherapy in 2021, with over 90% of the burden occurring in sub-Saharan Africa (WHO, 2023). The disease disproportionately affects rural and resource-limited communities where access to safe water and sanitation is limited.

In East Africa, schistosomiasis prevalence varies widely across and within countries due to differences in hydrology, land use, climate, and socio-economic conditions. High transmission zones are commonly reported around Lake Victoria, the Indian Ocean coastline, major river systems, and irrigation schemes. These variations demonstrate that schistosomiasis transmission is highly localized and spatially heterogeneous (WHO, 2023).

In Kenya, schistosomiasis remains endemic in the Lake Victoria basin, the Coastal region, and irrigation areas of Central Kenya. National surveys and localized studies consistently report marked differences in prevalence between counties, sub-counties, and neighboring communities, indicating that transmission dynamics are strongly influenced by local environmental and social conditions (WHO, 2023).

Schistosomiasis Among Children Aged 9–15 Years

Children aged 9–15 years are among the most vulnerable groups to schistosomiasis infection. At this developmental stage, children increasingly participate in activities such as swimming, fishing, herding, and water collection, which increase contact with contaminated

freshwater sources (WHO, 2023). In addition, partial immunity acquired earlier in childhood is often insufficient to prevent reinfection.

Studies conducted in Western Kenya have reported *Schistosoma mansoni* prevalence exceeding 70% among school-aged children living near Lake Victoria (Odiere et al., 2012). Furthermore, research in irrigation areas of Central Kenya has documented high prevalence among children due to frequent contact with irrigation canals that provide favorable habitats for intermediate snail hosts (Chieng et al., 2006).

Chronic schistosomiasis during late childhood and early adolescence has been associated with anaemia, stunted growth, impaired cognitive development, and reduced school attendance (WHO, 2023). These effects highlight the importance of focusing epidemiological research and control interventions on children within this age group.

Determinants Of Bilharzia Prevalence

Environmental Determinants

Environmental factors play a Central role in schistosomiasis transmission. Proximity to freshwater bodies such as rivers, lakes, irrigation canals, and ponds significantly increases the risk of infection, particularly in communities that rely on these water sources for daily activities (WHO, 2023).

In Central Kenya, Chieng et al. (2006) reported a schistosomiasis prevalence of 72.5% among school-aged children living in irrigation areas of Kirinyaga County. The high prevalence was attributed to frequent water contact and the presence of suitable habitats for intermediate snail hosts. However, the study did not examine the spatial distribution of infections across villages or administrative units.

In Western Kenya, geospatial modelling studies have demonstrated that schistosomiasis prevalence decreases with increasing distance from Lake Victoria, highlighting the spatial influence of hydrological features on disease distribution (Woodhall et al., 2013). These findings emphasize the importance of incorporating spatial location into epidemiological analysis.

Poor sanitation further sustains transmission by facilitating contamination of freshwater bodies with parasite eggs, leading to persistent local transmission and infection clustering (Coluzzi & Pignatti, 2004).

Demographic Determinants

Age is a well-established determinant of schistosomiasis prevalence, with infection rates increasing throughout childhood and peaking during adolescence due to cumulative exposure (WHO, 2023). Gender differences in prevalence have also been reported, often reflecting culturally defined roles in water-related activities such as fishing and water collection.

Socio-economic factors interact with demographic characteristics to shape infection risk. Studies conducted in Migori County found that children from households with limited access to safe water and sanitation facilities had significantly higher infection risk, contributing to geographically concentrated disease burden (Ng’ang’a et al., 2016).

Control Strategies

Mass Drug Administration

Mass drug administration using praziquantel remains the primary strategy for schistosomiasis control. The World Health Organization recommends periodic treatment of at-risk populations, particularly school-aged children, to reduce morbidity and interrupt transmission (WHO, 2023).

Despite repeated mass drug administration campaigns in Kenya, persistent transmission has been reported in several endemic areas. Studies from Western and Central Kenya indicate that reinfection often occurs shortly after treatment, suggesting that uniform treatment strategies may fail to address localized transmission hotspots.

Spatial Epidemiology and Statistical Approaches

Spatial epidemiology is a branch of epidemiology concerned with the study of geographical variation in disease incidence and prevalence, and the relationship between health outcomes and spatially distributed environmental, demographic, and socio-economic factors (UFZ, n.d.). It explicitly incorporates spatial location into analysis, recognizing that observations from nearby geographical areas may be more similar than those further apart.

Spatial epidemiological methods are used to identify disease clustering, spatial dependence, and geographical gradients. Techniques such as disease mapping, measures of spatial autocorrelation, and spatial regression models allow researchers to assess whether observed patterns are random or spatially structured.

In schistosomiasis research, spatial epidemiology is particularly relevant because transmission is environmentally driven and location-dependent. Failure to account for spatial autocorrelation may result in biased estimates and misleading statistical inference.

Research Gaps

Despite extensive research on schistosomiasis in Kenya, several important gaps remain. Many studies focus on estimating prevalence and identifying individual-level risk factors without analyzing the spatial distribution of infections. For example, Chieng et al. (2006) reported a prevalence of 72.5% among school-aged children in Kirinyaga County but did not examine spatial variation or clustering of cases. Similarly, studies conducted in Migori County

documented schistosomiasis prevalence and co-infection patterns but relied mainly on descriptive statistics, without applying spatial statistical techniques to identify geographical hotspots (Ng'ang'a et al., 2016).

Research in Mbita District reported high prevalence associated with proximity to Lake Victoria but did not formally test for spatial autocorrelation or model spatial dependence (Odiere et al., 2012). Consequently, few Kenyan studies integrate environmental, demographic, and spatial variables within a unified spatial statistical framework.

This study addresses these gaps by applying spatial statistical methods to analyse spatial variation, clustering, and determinants of Bilharzia prevalence among children aged 9–15 years in the Coastal region of Kenya.

Summary

This chapter has reviewed literature on schistosomiasis prevalence, determinants, and spatial distribution, with emphasis on school-aged children. The review highlights substantial geographical heterogeneity in disease burden and limited application of spatial statistical methods in Kenyan studies. These gaps justify the use of spatial epidemiology to inform targeted control strategies in the Coastal region of Kenya.

CHAPTER THREE: METHODOLOGY

Introduction

This chapter describes the methodology that was applied to examine the spatial distribution and spatial determinants of Bilharzia prevalence among children aged 9–15 years in the Coastal region of Kenya. The study adopted a spatial epidemiological approach, combining spatial descriptive analysis and spatial statistical techniques to assess geographical variation, spatial association, and clustering of Bilharzia prevalence across sub-counties. The methodology is designed to ensure that spatial patterns and relationships are systematically analyzed using appropriate statistical and geographic tools.

Study Area

The study was conducted in the Coastal region of Kenya, an area known to experience persistent transmission of Bilharzia due to favorable environmental and ecological conditions. The region is characterized by numerous rivers, ponds, irrigation schemes, and other freshwater bodies that facilitate the breeding of intermediate snail hosts. Variations in access to improved water sources and sanitation facilities across the region are expected to influence Bilharzia transmission. For the purpose of spatial analysis, the study area was subdivided into sub-counties, which served as the primary spatial units of analysis.

Study Design

A cross-sectional spatial study design was employed. This design involved the analysis of existing survey data collected at a single point in time and the aggregation of individual-level observations to the sub-county level. The design is appropriate for examining spatial variation and identifying geographical patterns of disease prevalence across administrative units (Gao et al., 2023).

Study Population

The study population will consist of children aged 9–15 years residing in the Coastal region of Kenya. This age group was selected because school-age children are known to have the highest exposure to Bilharzia infection due to frequent contact with contaminated water during domestic and recreational activities. Focusing on this population allowed the study to capture spatial patterns of infection within a high-risk group (Schistosomiasis, 2023).

Data Sources

The study used secondary data collected from the Coastal region of Kenya. The dataset includes individual-level information on Bilharzia infection status, age, gender, water source, and toilet availability. For spatial analysis, the individual-level data was aggregated to the sub-county level.

Geographical data in the form of sub-county boundary shapefiles was obtained from recognized geographic data repositories, such as the Global Administrative Areas (GADM) database. These shapefiles provided the geographic boundaries and spatial relationships necessary for spatial statistical analysis (GADM, 2024). https://gadm.org

Variables Of the Study

Dependent Variable

The dependent variable was Bilharzia prevalence, which was calculated at the sub-county level as the proportion of children aged 9–15 years who test positive for Bilharzia infection.

Independent Variables

All independent variables were spatially aggregated at the sub-county level:

- **Demographic Variables**:
    - Mean age of children per sub-county

    - Proportion of male and female children per sub-county

- **Environmental Variables**:
    - Proportion of children using unprotected water sources per sub-county
    - Proportion of children without access to toilet facilities per sub-county

These variables were selected based on their documented association with Bilharzia transmission and their relevance to spatial variation in infection risk (Kepha et al., 2023).

Data Management

Data management involved checking the dataset for missing values, inconsistencies, and outliers. Individual-level records will be aggregated to compute sub-county-level summaries, including Bilharzia prevalence and proportions of demographic and environmental characteristics. The aggregated dataset was then linked to the sub-county shapefiles using unique geographic identifiers to ensure accurate spatial alignment.

Data Analysis

Spatial Descriptive Analysis

Spatial descriptive analysis was conducted to examine the geographical distribution of Bilharzia prevalence across sub-counties, and was calculated as,

$$P_i = \frac{c_i}{N_i} \times 100 \qquad (1)$$

where P_i = Bilharzia prevalence in sub-county i, C_i = number of infected children in sub-county i, and N_i = total number of children examined in sub-county i.

The mean prevalence across all sub-counties was expressed as,

$$\bar{P} = \frac{1}{n}\sum_{i=1}^{n} P_i \qquad (2)$$

To test for the variation of prevalence of Bilharzia across all sub-counties, I measured it using the standard deviation formula.

$$s = \sqrt{\frac{\sum_{i=1}^{n}(P_i - \bar{P})^2}{n-1}} \qquad (3)$$

where s represents the standard deviation of Bilharzia prevalence.

Choropleth maps were generated to visually display variation in prevalence levels and to identify high-prevalence and low-prevalence sub-counties.

$Z_{(S_i)} = P_i$, where $Z_{(S_i)}$ represents the value of Bilharzia prevalence at spatial location s_i.

Mapping was providing an initial understanding of spatial patterns and guides further spatial statistical analysis (Lin et.al., 2022).

Spatial Autocorrelation Analysis

Global spatial autocorrelation was assessed using Moran's I statistic to determine whether Bilharzia prevalence across sub-counties is spatially clustered, dispersed, or randomly distributed. Positive Moran's I values indicate spatial clustering, negative values indicate spatial dispersion, and values close to zero indicate random spatial distribution.

$$I = \frac{n}{W}\frac{\sum_{i=1}^{n}\sum_{j=1}^{n} w_{ij}(P_i - \bar{P})(P_j - \bar{P})}{\sum_{i=1}^{n}(P_i - \bar{P})^2} \qquad (4)$$

where:

n represents the number of sub-counties (spatial units)

P_i represents prevalence in the sub-county i

P_j represents prevalence in the neighboring sub-county j

$\bar{P}$ is the mean prevalence

w_{ij} are the spatial weights between locations i and j.

Moran's I was computed based on a spatial weights matrix that defines neighboring sub-counties. The spatial weights matrix W will define the neighborhood structure between sub-counties

$$w_{ij} = \begin{cases} 1, if\ sub-county\ i\ and\ j\ share\ a\ boundary \\ 0, otherwise \end{cases} \quad (5)$$

A statistically significant Moran's I value indicates the presence of spatial dependence in Bilharzia prevalence (Anselin, 1995).

Spatial Association Analysis

To examine spatial determinants of Bilharzia prevalence, spatial regression analysis,

$$Y_i = \beta_0 + \beta_1 X_{1i} + \beta_2 X_{2i} + \beta_3 X_{3i} + \beta_4 X_{4i} + \varepsilon_i \quad (6)$$

where:

Y_i is the Bilharzial prevalence in sub-county i

X_1 is the mean age

X_2 is the gender proportion

X_3 is the water source

X_4 is the toilet availability

β_0 is intercept

$\beta_1 ... \beta_4$ are regression coefficients

ε_i is the error term,

was conducted using sub-county–level aggregated data. Bilharzia prevalence was modelled as a function of spatially aggregated demographic and environmental variables. This analysis assessed how variation in these factors across geographical space is associated with variation in Bilharzia prevalence, while maintaining a purely spatial unit of analysis (Lun et al., 2022).

Software Used

All statistical and spatial analyses was conducted using Excel and R statistical software. Spatial data handling, mapping, and spatial autocorrelation analysis was performed using appropriate R packages designed for spatial analysis.

Ethical Considerations

The study relied on secondary data that does not contain personal identifiers. Ethical approval and permission for data use was obtained by the original data collectors. Confidentiality and anonymity of study participants have been maintained throughout the study.

CHAPTER FOUR: DATA ANALYSIS

Data Cleaning

The initial data, which I had obtained from a research doctor, had 31275 children's entries with ages ranging from 8 to 15, with two 99 and 999 representing unrecorded age. I eliminated all other ages and remained with 9 to 15 years old only and omitted records with missing values. Afterwards I added longitude and longitude variables from the from Kenya - Subnational Administrative Boundaries to align with my spatial statistics. Therefore, the remaining variables of my interest were, age, gender, county, subcounty, availability of toilet and treated water, and the total children remaining was 26,118.

Overall Bilharzia Prevalence

In R, I did various analysis with 26,118 children aged 9-15 years after all the cleaning. The overall prevalence of bilharzia was found to be 6.34% which aligns closely with national estimates for the Coastal region and confirms that despite ongoing control efforts, it remains a moderate health burden among this high-risk age group.

Spatial Variation in Bilharzia Prevalence Across Sub-counties (objective 1)

The prevalence exhibited marked spatial heterogeneity across all the sub-counties, as shown in Table 4.1

Table 4.1: Bilharzia prevalence by sub-counties

County	**Sub-county**	**Total children**	**Infected**	**Prevalence (%)**
Tana river	Galole	797	234	29.4
Kilifi	Magarini	1613	308	19.1
Tana River	Garsen	1034	172	16.6
Kilifi	Rabai	875	87	9.94
Kwale	Msambweni	775	71	9.16
Kwale	Matuga	1097	99	9.02
Tana River	Bura	864	73	8.45
Kwale	Lunga Lunga	879	74	8.42
Lamu	Lamu West	1068	86	8.05
Taita Tateva	Taveta	1154	68	5.89
Kilifi	Kaloleni	926	49	5.29
Kilifi	Kilifi South	878	45	5.13
Kwale	Kinango	1414	70	4.95
Kilifi	Malindi	854	38	4.45
Taita Taveta	Voi	1228	41	3.34

Mombasa	Kisauni	1430	43	3.01
Kilifi	Ganze	834	15	1.80
Mombasa	Nyali	1116	20	1.79
Mombasa	Likoni	1063	19	1.79
Mombasa	Changamwe	1156	16	1.38
Mombasa	Jomvu	720	9	1.25
Lamu	Lamu East	140	1	0.714
Kilifi	Kilifi North	1446	8	0.553
Mombasa	Mvita	969	4	0.413
Taita Taveta	Wundanyi	763	3	0.393
Taita Taveta	Mwatate	1025	2	0.195

.

Statistical Test for Spatial Variation (objective 2)

Pearson's Chi-squared test with simulated p-value (based on 2000 replicates) test for independence between sub-county and infection status yielded:

$x^2 = 1900.2$, p = 0.0004998.

The null hypothesis (H_{01}: no significant spatial variation) is rejected at $p<0.001$. This is evident since the prevalence shows highly significant spatial variation with clear hotspots in Tana River (Galole and Garsen) and northern Kilifi Sub-counties. The prevalence also ranges

from a high of 29.4% to a low of 0.195%, representing a 150-fold difference. This confirms the need for spatially targeted interventions.

Spatial Distribution of Demographic Characteristics and Association with Prevalence (objective 3)

Table 4.2: Demographic characteristics by sub-counties

Sub-county	**Prop_male**	**Mean_age**	**Prevalence (%)**
Galole	0.519	10.9	29.4
Magarini	0.507	11.4	19.1
Garsen	0.493	11.0	16.6
Rabai	0.530	11.0	9.94
Msambweni	0.515	11.2	9.16
Matuga	0.524	11.4	9.02
Bura	0.483	11.0	8.45
Lunga Lunga	0.466	11.5	8.42
Lamu West	0.532	11.3	8.05
Taveta	0.515	11.2	5.89
Kaloleni	0.498	11.4	5.29
Kilifi South	0.497	11.1	5.13
Kinango	0.504	11.2	4.95
Malindi	0.501	11.2	4.45
Voi	0.507	11.3	3.34

Kisauni	0.495	11.2	3.01
Ganze	0.463	11.3	1.80
Nyali	0.499	11.3	1.79
Likoni	0.468	11.3	1.79
Changamwe	0.492	11.3	1.38
Jomvu	0.467	11.2	1.25
Lamu East	0.407	11.3	0.714
Kilifi North	0.515	11.4	0.553
Mvita	0..513	11.1	0.413
Wundanyi	0.503	11.4	0.393
Mwatate	0.513	11.2	0.195

Individual Level Association

Gender

- Pearson's Chi-square test with Yates' continuity correction showed a highly significant association:

$\chi^2 = 22.818$, df = 1, $p = 1.781 \times 10^{-6}$. Boys had significantly higher infection rates than girls.

Age

- Welch’s two sample t-test comparing the mean age between infected and non-infected children:

t = -0.601, df = 1887.1, p = 0.5479 (95% CI: -0.109 to 0.058). Mean age was virtually identical, that is, 11.23 years uninfected vs. 11.26 years infected.

Aggregate Level Associations

- Proportion of males vs. prevalence: Pearson correlation r = 0.308, p = 0.1256 (statistically significant)
- Mean age vs prevalence: Pearson correlation r = -0.423, p = 0.03129 (statistically significant) sub-counties with slightly younger average age tended to report higher prevalence.

The null hypothesis (H_{02}: no significant spatial association between demographic and prevalence) is rejected. As per the results, gender is a strong individual-level risk factor with boys being at a higher risk due to likely greater contacts with water bodies behavior. This suggests the importance of gender-sensitive messaging in health education campaigns.

Spatial Association Between Environmental Factors and Prevalence (objective 4)

Table 4.3: Environmental factors by sub-counties

Sub-county	**Prop_water_1**	**Propp_toilet_1**	**Prevalence (%)**
Galole	0.432	0.605	29.4
Magarini	0.479	0.551	19.1
Garsen	0.688	0.544	16.6
Rabai	0.639	0.917	9.94
Msambweni	0.822	0.711	9.16
Matuga	0.553	0.824	9.02

Bura	0.302	0.719	8.45
Lunga Lunga	0.382	0.650	8.42
Lamu West	0.723	0.776	8.05
Taveta	0.697	0.963	5.89
Kaloleni	0.294	0.826	5.29
Kilifi South	0.743	0.925	5.13
Kinango	0.330	0.357	4.95
Malindi	0.934	0.690	4.45
Voi	0.852	0.977	3.34
Kisauni	0.931	0.981	3.01
Ganze	0.650	0.683	1.80
Nyali	0.861	0.996	1.79
Likoni	0.675	0.949	1.79
Changamwe	0.881	0.997	1.38
Jomvu	0.6	0.938	1.25
Lamu East	0.25	1	0.714
Kilifi North	0.953	0.860	0.553
Mvita	0.745	0.980	0.413
Wundanyi	0.461	0.999	0.393
Mwatate	0.72	0.995	0.195

Individual-Level Associations

- Water source: Pearson's chi-square test: $\chi^2 = 213.82$, df = 1, $p < 2.2 \times 10^{-16}$.

Children using unsafe water sources had dramatically higher infection rates.

- Toilet availability: Pearson's chi-square test: $\chi^2 = 341.47$, df = 1, $p < 2.2 \times 10^{-16}$.

Absence of a toilet was strongly associated with infection

Aggregate Level Associations

- Proportion using Water_Source = 1 vs. prevalence: Pearson correlation r = –0.290, p = 0.1501 (not significant).
- Proportion with Toilet = 1 vs. prevalence: Pearson correlation r = –0.614, p = 0.0008443 (highly significant). Sub-counties with higher toilet coverage had substantially lower prevalence.

The null hypothesis (H_{03}: no significant spatial association between environmental factors and prevalence) is rejected. Both water source and toilet availability are strongly linked to infection at the individual level, while toilet coverage emerges as the dominant protective factor at the sub-county level. These results provide clear evidence for WASH (Water, Sanitation, and Hygiene) interventions as a core control strategy.

Presence and Strength of Spatial Autocorrelation in Bilharzia Prevalence (objective 4)

I performed Global Moran's I test (5-nearest neighbors spatial weights matrix) on sub-county prevalence rates and the results were;

- Moran's I statistic = 0.235 (positive spatial autocorrelation), Standard deviate = 3.2926, and p-value = 0.0004962.

The null hypothesis (H_{04}: no significant spatial autocorrelation) is rejected at $p < 0.001$. Bilharzia prevalence exhibits moderate but highly significant positive spatial clustering. High-prevalence sub-counties tend to be geographically proximate (notably in Tana River and northern Kilifi), while low-prevalence areas also cluster together. This validates the use of spatial statistics for targeted resource allocation.

Multivariate Risk Models

I used two multivariate models to determine independent effects while controlling for confounding: multivariate logistic regression (individual level) and multivariate linear regression (sub-county aggregate level).

Multivariate Logistic Regression

- Outcome: binary infection status
- Predictors: Gender, Age, Water_Source, Toilet.
- Model fit: Null deviance 12335, residual deviance 11909(AIC=11919)
- Likelihood ratio test $p < 0.001$

Table 4.4: Adjusted odds ratios for multivariate logistic regression

Variable	**Adjusted OR**	**95% CI**	**p-value**	**Interpretation**
Gender (male vs female)	1.291	1.167-1.429	7.38×10^{-7}	Boys have 29.1% higher odds(independent)

Age (per year)	1.015	0.985-1.046	0.326	No independent effect
Water_Source (1 vs 0)	0.575	0.518-0.638	$< 2.2 \times 10^{-16}$	Safe water reduces odds by 42.5%
Toilet (1 vs 0)	0.442	0.396-0.0494	$< 2.2 \times 10^{-16}$	Toilet reduces odds by 55.8% (strongest protector)

Multivariate Linear Regression

- Outcome: prevalence
- Predictors: prop_male, mean_age, prop_water_1, Prop_toilet_1.
- Model fit: adjusted $R^2 = 0.4738$ (explains 47.4% of spatial variation in prevalence). Overall $F = 6.628$, $p = 0.001298$.

Table 4.5: Coefficients for multivariate linear regression

Predictor	$\boldsymbol{\beta}$	**Std.error**	**t-value**	**p-value**	**Interpretation**
prop_male	0.76481	0.42229	1.811	0.0845	Marginally positive
Mean_age	-0.13057	0.06769	-1.929	0.0674	Marginally negative

prop_water_1	-0.07435	0.05494	-1.353	0.1903	Not significant after adjustments
Prop_toilet_1	-0.16988	0.06194	-2.743	0.0122	Strongest independent factor

Interpretation of Risk Factors

- Toilet availability is the single most robust protective factor at both individual (OR 0.442) and ecological (β = –0.16988) levels.
- Unsafe water source has a strong direct effect but shares variance with sanitation.
- Male gender remains an independent behavioral risk factor.
- Age shows no independent influence.
- The multivariate models confirm that WASH-related factors explain more unique variance than demographics, reinforcing the spatial hotspots identified in Objective 1.

CHAPTER FIVE: INTERPRETATION, CONCLUSION AND RECOMMENDATIONS

Interpretation of Key Findings

The results of this study provide compelling evidence that Bilharzia (schistosomiasis) in Kenya's Coastal region is not randomly distributed but exhibits strong spatial patterning, demographic disparities, and clear environmental drivers. These findings align closely with the background presented in Chapter One and the literature reviewed in Chapter Two, which highlighted geographical variation, clustering in high-risk areas, and the critical role of water-contact behaviors and sanitation among school-age children (Kepha et al., 2023; Mutono et al., 2025).

Spatial Variation and Clustering (Objective 1 & 4)

The extreme spatial heterogeneity (prevalence ranging from 29.4% in Tana River–Galole to 0.195% in Taita Taveta–Mwatate) and significant positive spatial autocorrelation (Moran's $I = 0.235$, $p = 0.0005$) confirm that infection is geographically clustered. High-prevalence sub-counties (Galole, Magarini, Garsen, Rabai) are not isolated; they form contiguous hotspots, consistent with the ecological requirements of intermediate snail hosts and frequent human-water contact in riverine and irrigation areas. This clustering has important implications: interventions in one high-burden sub-county are likely to produce spillover benefits to neighboring areas. The results strongly support the precision mapping approach advocated in the project document and echo national calls for sub-county-level targeting rather than blanket county-wide programs.

Demographic Risk Factors (Objective 2)

Gender emerged as a robust independent risk factor (adjusted OR = 1.291, $p < 0.001$), with boys at significantly higher risk even after controlling for age, water source, and toilet access. This is almost certainly behavioral: boys engage more frequently in swimming, fishing, and playing in infested water bodies. Age showed no independent individual-level effect (OR = 1.015, $p = 0.326$) but a modest negative ecological association ($r = –0.423$, $p = 0.031$), suggesting that sub-counties with slightly younger average ages experience marginally higher transmission pressure. These demographic patterns reinforce the need for gender-sensitive health education and confirm that school-age children remain the highest-risk group, as stated in the study justification.

Environmental Risk Factors and Multivariate Insights (Objective 3)

The strongest and most consistent findings concern environmental determinants. At the individual level, lack of a toilet (adjusted OR = 0.442, $p < 2.2 \times 10^{-16}$) and unsafe water source (adjusted OR = 0.575, $p < 2.2 \times 10^{-16}$) were highly significant. At the sub-county level, toilet coverage remained the only significant independent predictor in the multivariate linear model ($\beta = –0.16988$, $p = 0.0122$), explaining nearly half the spatial variation in prevalence (Adjusted $R^2 = 0.4738$). The multivariate models are particularly powerful because they demonstrate independent effects: toilet availability retains its protective role even after accounting for water source and gender. This indicates that sanitation is not merely a proxy for poverty or water access but a direct transmission barrier (by reducing contamination of freshwater bodies with parasite eggs).

Implications for Interventions

The results provide a clear roadmap for evidence-based, spatially targeted control. Interventions must be integrated, multi-level, and community-owned to achieve sustainable reduction.

Interventions with County and National Authorities

1. Precision Targeting of Hotspots

- Prioritize mass drug administration (MDA) with praziquantel in the top 8 high-prevalence sub-counties (prevalence >8%): Tana River (Galole, Garsen, Bura) and Kilifi (Magarini, Rabai).
- Use the sub-county prevalence map generated in this study to guide annual MDA rounds and integrate it into the Kenya National Neglected Tropical Diseases Strategic Plan.
- Allocate resources proportionally: sub-counties with prevalence >15% should receive at least twice the MDA coverage of low-prevalence areas.

2. WASH Infrastructure Scale-up

- Launch a targeted sanitation campaign in hotspot sub-counties aiming for >90% toilet coverage within 3–5 years.
- Partner with the Ministry of Health, Ministry of Water & Sanitation, and county governments to construct community and school latrines using low-cost, climate-resilient designs.
- Introduce safe water points (boreholes or treated piped water) in areas where Water_Source = 0 is still dominant.

3. Surveillance and Monitoring

- Establish sub-county-level sentinel surveillance using the same ChildID-linked dataset format to track prevalence annually.
- Incorporate spatial autocorrelation metrics (Moran's I) into routine reporting so authorities can detect emerging clusters early.

4. Policy Integration

- Embed the multivariate findings into county Integrated Development Plans (CIDPs) and the National School Health Program.
- Advocate for ring-fenced funding for Coastal schistosomiasis control under the Kenya NTD Master Plan.

Interventions Involving Residents and Communities

1. Behavior Changes Communication (BCC)

- Design gender-specific messages: target boys with campaigns on "safe water play" and avoiding swimming in high-risk rivers/ponds.
- Use local radio, school clubs, and community health volunteers (CHVs) to promote consistent toilet use and handwashing.
- Involve religious and traditional leaders in hotspot villages to endorse sanitation and reduce open defecation.

2. Community-Led Total Sanitation (CLTS)

- Implement CLTS in the 10 highest-prevalence sub-counties, empowering residents to build and maintain their own latrines.
- Reward “Open Defecation Free” villages with recognition and small incentives (e.g., school supplies).

3. School-Based Programs

- Integrate Bilharzia education into the school curriculum with practical sessions on water safety and toilet maintenance.
- Establish student health clubs that monitor local water points and report unsafe sources to authorities.

4. Economic and Livelihood Support

- Link WASH interventions with livelihood programs (e.g., fish farming in safe ponds or irrigation improvements) to reduce dependence on infested natural water bodies.

Integrated Intervention Framework

- Short-term (0–12 months): Intensified MDA + emergency latrine construction in hotspots.
- Medium-term (1–3 years): Full WASH infrastructure + BCC campaigns.
- Long-term (3–5 years): Sustained surveillance, behaviors maintenance, and economic diversification.

This multi-pronged approach (MDA + WASH + behavior change) has proven effective in other African settings and is now strongly supported by the empirical evidence from this study

Study Limitations

- The analysis is cross-sectional; causal inference is strengthened by multivariate adjustment but longitudinal data would be ideal.
- Environmental variables were limited to water source and toilet type; future studies should incorporate snail density, water pH, and temperature.
- Spatial analysis used point centroids; ward-level shapefiles would allow finer-scale choropleth mapping.

Recommended Future Research:

- Spatially explicit multivariate models (e.g., Bayesian CAR models).
- Cost-effectiveness analysis of toilet-focused vs. MDA-only interventions.
- Qualitative studies exploring barriers to toilet adoption among residents.

Conclusion and Policy Recommendations

This study demonstrates that Bilharzia in Coastal Kenya is a spatially clustered, environmentally driven disease with clear modifiable risk factors—most notably lack of toilet access. By rejecting all four null hypotheses and quantifying independent effects through multivariate modelling, the findings provide actionable intelligence for authorities and residents alike. Key policy recommendation: Shift from uniform MDA to a precision public health strategy that combines targeted chemotherapy with aggressive sanitation improvement in the identified hotspots. Collaboration between national government, county authorities, and communities is essential for success. Implementing the interventions outlined above has the

potential to reduce prevalence by more than 50% within five years in the worst-affected sub-counties, aligning with Kenya's commitment to eliminate schistosomiasis as a public health problem.

References

Anselin, L. (1995). Local indicators of spatial association—LISA. *Geographical Analysis*, *27*(2), 93–115. https://doi.org/10.1111/j.1538-4632.1995.tb00338.x

Chacha, G. (2025, May 15). *Study: Suckling children at risk of Bilharzia too*. Health. https://www.standardmedia.co.ke/health/health-science/article/2001519158/study-suckling-children-at-risk-of-Bilharzia-too

Chieng, A.O., Mutuku, M.W., Muhoho, N.D., et al. (2006). Prevalence and intensity of *Schistosoma mansoni* in irrigation and non-irrigation areas of Central Kenya. *Transactions of the Royal Society of Tropical Medicine and Hygiene*, 100(5), 422–430. https://pubmed.ncbi.nlm.nih.gov/16463753/

Colley, D.G., Bustinduy, A.L., Secor, W.E., & King, C.H. (2014). Human schistosomiasis. *The Lancet*, 383(9936), 2253–2264. https://pubmed.ncbi.nlm.nih.gov/24698483/

Coluzzi, M., & Pignatti, M. (2004). Environmental and ecological determinants of schistosomiasis transmission. *Transactions of the Royal Society of Tropical Medicine and Hygiene*, 98(9), 491–498. https://pubmed.ncbi.nlm.nih.gov/15457271/

Gao, B., Yang, J., Chen, Z., Sugihara, G., Li, M., Stein, A., ... & Wang, J. (2023). Causal inference from cross-sectional earth system data with geographical convergent cross mapping. *Nature Communications*, *14*(1), 5875. https://doi.org/10.1038/s41467-023-43055-y

Kepha, S., Ochol, D., Wakesho, F., Omondi, W., Njenga, S. M., Njaanake, K., Kihara, J., Mwatha, S., Kanyi, C., Oloo, J. O., Kibati, P., Yard, E., Appleby, L. J., McRae-

McKee, K., Odiere, M. R., & Matendechero, S. H. (2023). undefined. *PLOS Neglected Tropical Diseases*, *17*(1), e0011043. https://doi.org/10.1371/journal.pntd.0011043

Lin, C., & Wen, T. (2022). How spatial epidemiology helps understand infectious human disease transmission. *Tropical Medicine and Infectious Disease*, *7*(8), 164. https://doi.org/10.3390/tropicalmed7080164

Lun, X., Wang, Y., Zhao, C., Wu, H., Zhu, C., Ma, D., Xu, M., Wang, J., Liu, Q., Xu, L., & Meng, F. (2022). Epidemiological characteristics and temporal-spatial analysis of overseas imported dengue fever cases in outbreak provinces of China, 2005–2019. *Infectious Diseases of Poverty*, *11*(1). https://doi.org/10.1186/s40249-022-00937-5

Mutono, N., Njuguna, B., Kepha, S., Wakesho, F., Omondi, W., Kibati, P., Chami, I., Ower, A., McRae-McKee, K., Juma, C., Sempele, I., Ouma, V., Muchangi, J. M., Odiere, M. R., & Thumbi, S. (2025). Geostatistical modelling of soil-transmitted helminth prevalence in Kenya: Informing targeted interventions to accelerate elimination efforts. *International Journal of Infectious Diseases*, *156*, 107916. https://doi.org/10.1016/j.ijid.2025.107916

Ng'ang'a, M.M., Matendechero, S.H., Kariuki, L., & Omedo, M. (2016). Spatial distribution and co-infection with urogenital and intestinal schistosomiasis among primary school children in Migori County, Kenya. *East African Medical Journal*, 93(9), 450–458. https://erepository.uonbi.ac.ke/handle/11295/100448

Odiere, M.R., Rawago, F.O., Ombok, M., et al. (2012). High prevalence of schistosomiasis in Mbita and its adjacent islands of Lake Victoria, Western Kenya. *Parasites & Vectors*, 5, 278. https://parasitesandvectors.biomedCentral.com/articles/10.1186/1756-3305-5-278

Schistosomiasis. (2023, February 1). World Health Organization (WHO). https://www.who.int/news-room/fact-sheets/detail/schistosomiasis

UFZ – Helmholtz Centre for Environmental Research. (n.d.). Spatial epidemiology and environmental health applications. https://www.ufz.de/sentinel/index.php?en=46486

Woodhall, D.M., Wiegand, R.E., Wellman, M., et al. (2013). Use of geospatial modelling to predict *Schistosoma mansoni* prevalence in Nyanza Province, Kenya. *PLoS One*, 8(8), e71635. https://pubmed.ncbi.nlm.nih.gov/23922868

World Health Organization (WHO). (2023). Schistosomiasis: Fact sheet. https://www.who.int/news-room/fact-sheets/detail/schistosomiasis

Appendices

Work plan

Activity	Time
Introduction	1 week
Literature review	1 week
Methodology	2 weeks
Data acquisition and preparation	3 days
Data cleaning and management	1 week
Spatial data analysis	3 weeks
Interpretation of results	3 days
Final editing and submission	2 weeks

Budget

Printing	2000
Internet access	1000
Stationery	300
Miscellaneous	500
Total	3800